

NIT2113 – Cloud Application Development

Assessment 4: AWS Cloud Deployment Project – Smart Retail Website

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CloudFront URL:

<https://d25v3j5mr9xson.cloudfront.net>

S3 Website URL:

<https://ass4-megacart-web-bucket.s3.ap-southeast-2.amazonaws.com/index.html>

SNS Topic ARN:

<arn:aws:sns:ap-southeast-2:494186462539:megacart-promos>

Word Count (max 2000): 1928

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1 Technical Context and Project Goal

Scenario

For this project, we have been hired by a Smart Retail Solutions, Megacart, an Australian online retail business and asked to design a cloud-based promotional system because they rely entirely on digital marketing to reach new customers and promote new deals.

Goal

In order to achieve our clients needs, the goal of this project:

Design, build and deliver a secure website and working SNS topic with multiple subscribers to receive promotional notification

Success Criteria

To achieve the project goal, the deliverables required are:

- Public static site accessible via CloudFront with HTTPS and index.html as root
- SNS topic with at least 2 verified subscribers to receive messages
- Clear documentation and screenshots

2 Partner Contributions

Kevan Ly (S8126233):

My contribution for this project, I have setup the report and documentation of our implementation which I am responsible for the S3 bucket website hosting, generating website setting up the collaboration in AWS. In the report, I have contributed the Architecture Overview, Challenges and Solutions and Reflection of AI assistance. I have made additional improvements of the project by including CI/CD pipelines

Justin Dang (S8126233):

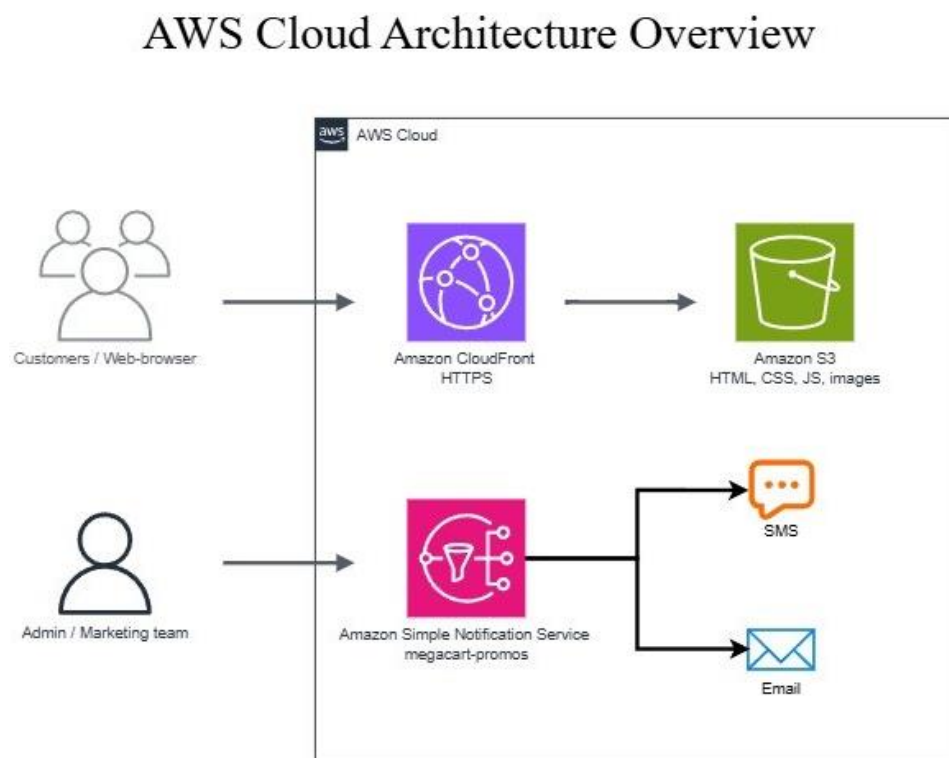
My contribution to this project included configuring Amazon CloudFront for secure website delivery with HTTPS support and connecting the CloudFront distribution to the Amazon S3 static website bucket. In addition, I assisted with configuring public access permissions and ACL settings for the S3 website hosting.

I was responsible for creating the SNS topic, adding and verifying subscribers, publishing promotional messages, and testing successful message delivery through email notifications. I also checked and documented the received SNS messages as part of the testing process.

For the report documentation, I completed Section 4 (Implementation), Section 6 (Justification of Tools including AI Assistant), and Section 8 (Optional Improvements / Next Steps). I also cooperated in completing Section 5 (Testing and Evidence) and Section 7 (Challenges and Solutions). Doing the report of 8.1. CI/CD Pipeline Automation.

3 Architecture Overview

Figure 1. AWS Cloud Architecture Overview



Cloud architecture showing Amazon S3, CloudFront, and SNS integration for the MegaCart promotional system.

This is an AWS cloud architecture overview diagram where the customer can access the website hosted from Amazon S3 through CloudFront that provides and improve the security of providing of HTTPS and delivering contents.

The notification side of the cloud architecture, Amazon SNS is managed and sent to subscribers via SMS and/or Email by admins or marketing teams to distribute promotional offers if customers have subscribed to megacart-promos.

4 Implementation

4.1 Development Environment and AI assistant

The chosen cloud IDE we chose was GitHub that uses a Copilot AI assistant generating HTML to create the structure, CSS to provide professional styling and JS for interactive functionality; files to build a standard website to include pages, products, promotional banner offers and additional elements and features to make the website presentable for customers. The reason to use this cloud IDE as it allows you to preview the codes and website in a separate console to allow the developer to make changes and display what the website will present to their customers. Additionally, the process was straightforward, simple and allow to streamline the project and has appropriate code structure.

4.2 Phase 1 – S3 Static Website

The first phase of the project focused on hosting the MegaCart promotional website using Amazon S3 static website hosting. The website files generated from GitHub Copilot, including HTML, CSS, JavaScript, and image assets, were uploaded into an Amazon S3 bucket configured for static website hosting.

After creating the S3 bucket, public access settings were adjusted to allow customers to access the website through the internet. Static website hosting was enabled within the bucket properties and the default root object was configured as index.html. Bucket policies and object permissions were then configured to allow public read access for website files.

During implementation, the team encountered an issue where the website displayed only unformatted text because some bucket objects were not publicly accessible. This issue was resolved by enabling public permissions for the remaining website objects inside the bucket.

After configuration was completed, the S3 website endpoint URL was tested successfully to verify that the promotional website was publicly accessible and fully functional.

4.3 Phase 2 – CloudFront (Secure, Scalable Delivery)

The second phase involved configuring Amazon CloudFront to securely distribute the S3-hosted website through HTTPS delivery. A CloudFront distribution was created and connected to the Amazon S3 bucket as the origin source.

The distribution was configured to redirect HTTP requests to HTTPS to improve website security and meet the assignment requirement for secure delivery. Additionally, the default root object was configured as index.html to ensure the website homepage loaded correctly when users accessed the CloudFront URL.

CloudFront improved the scalability and performance of the website by caching website content across AWS edge locations, allowing users to access website content more quickly from geographically closer servers. After deployment, the CloudFront domain URL was tested successfully to confirm secure website access and correct website functionality.

4.4 Phase 3 – Amazon SNS Promotional Notifications

The final phase focused on implementing a promotional notification system using Amazon SNS. A Standard SNS topic named megacart-promos was created to simulate marketing notifications for customers.

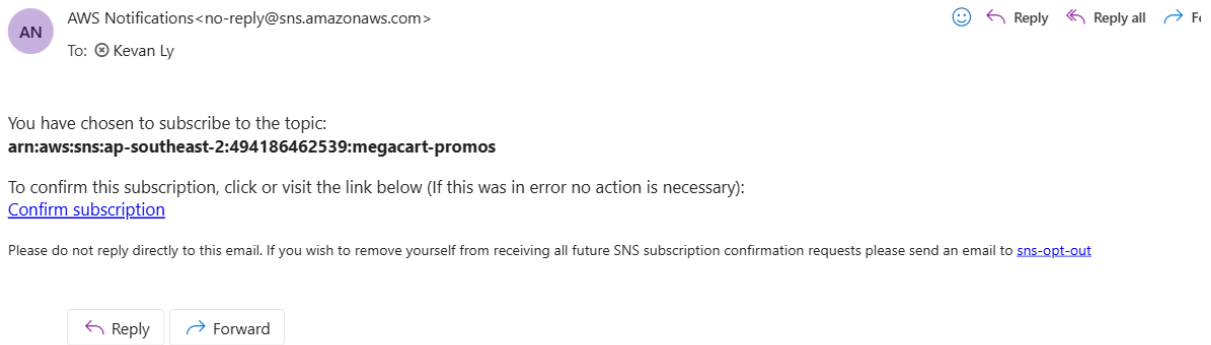
After creating the correct SNS topic, email subscriptions were added and verified through confirmation emails sent by AWS SNS. Once verification was completed, promotional messages were published successfully through the SNS topic.

A sample promotional message stating “New Deals at MegaCart! 30% off all electronics. Limited time only!” was sent to all verified subscribers. Successful delivery of the promotional notifications was confirmed through received email messages and documented through screenshots for testing evidence.

5 Testing and Evidence

Screenshot of Website with HTTPS URL, SNS receive message and confirmations,

Figure 2. SNS Subscription Confirmation Email



Verification email sent by Amazon SNS to confirm subscriber registration.

Figure 3. Successful SNS Subscription Confirmation

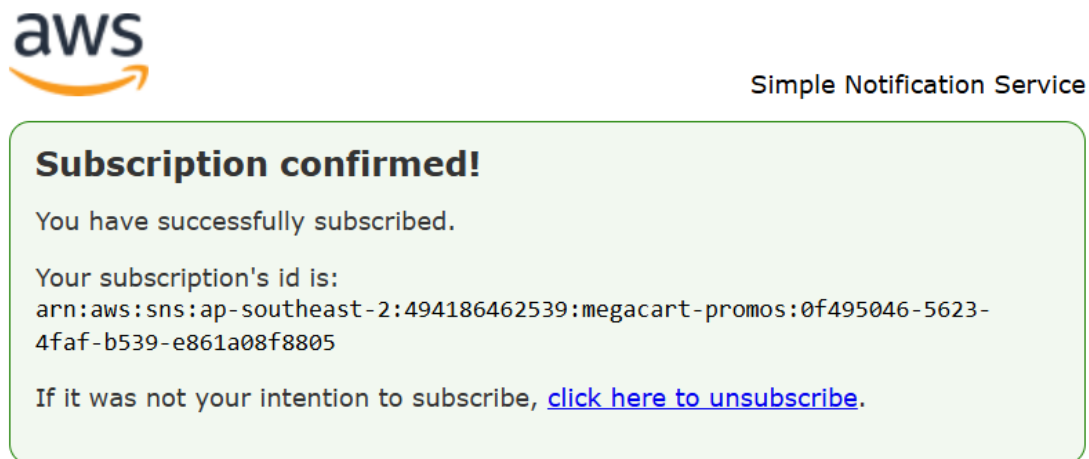
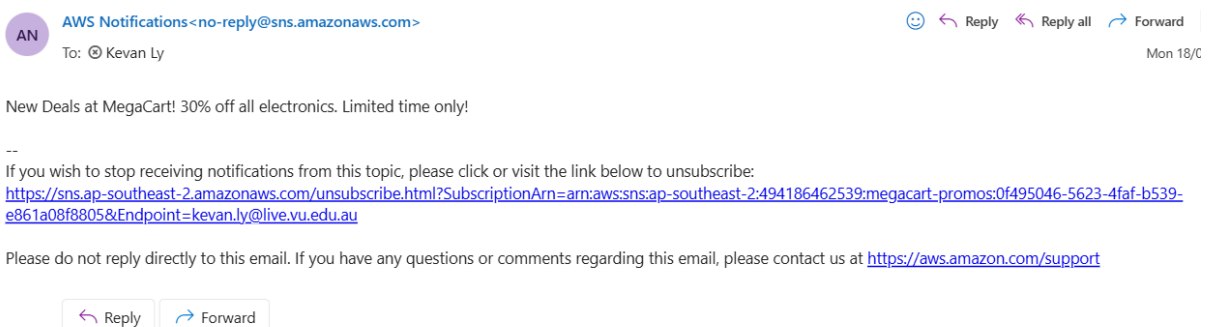
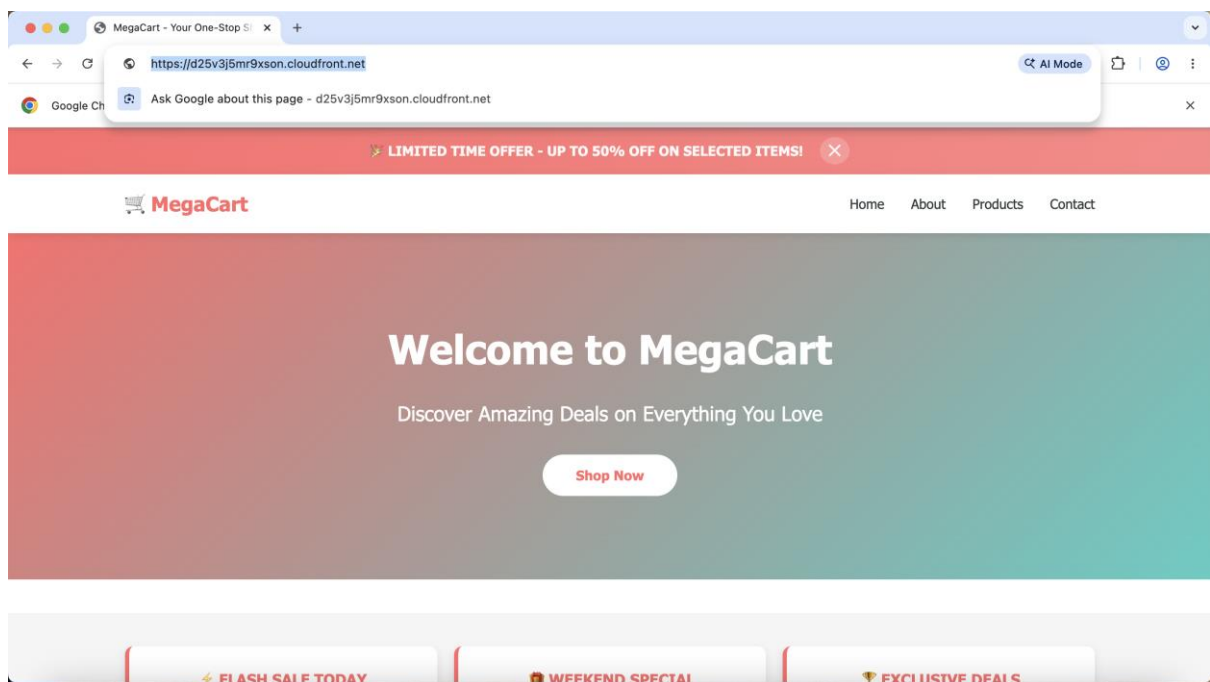


Figure 4. SNS Promotional Notification Email



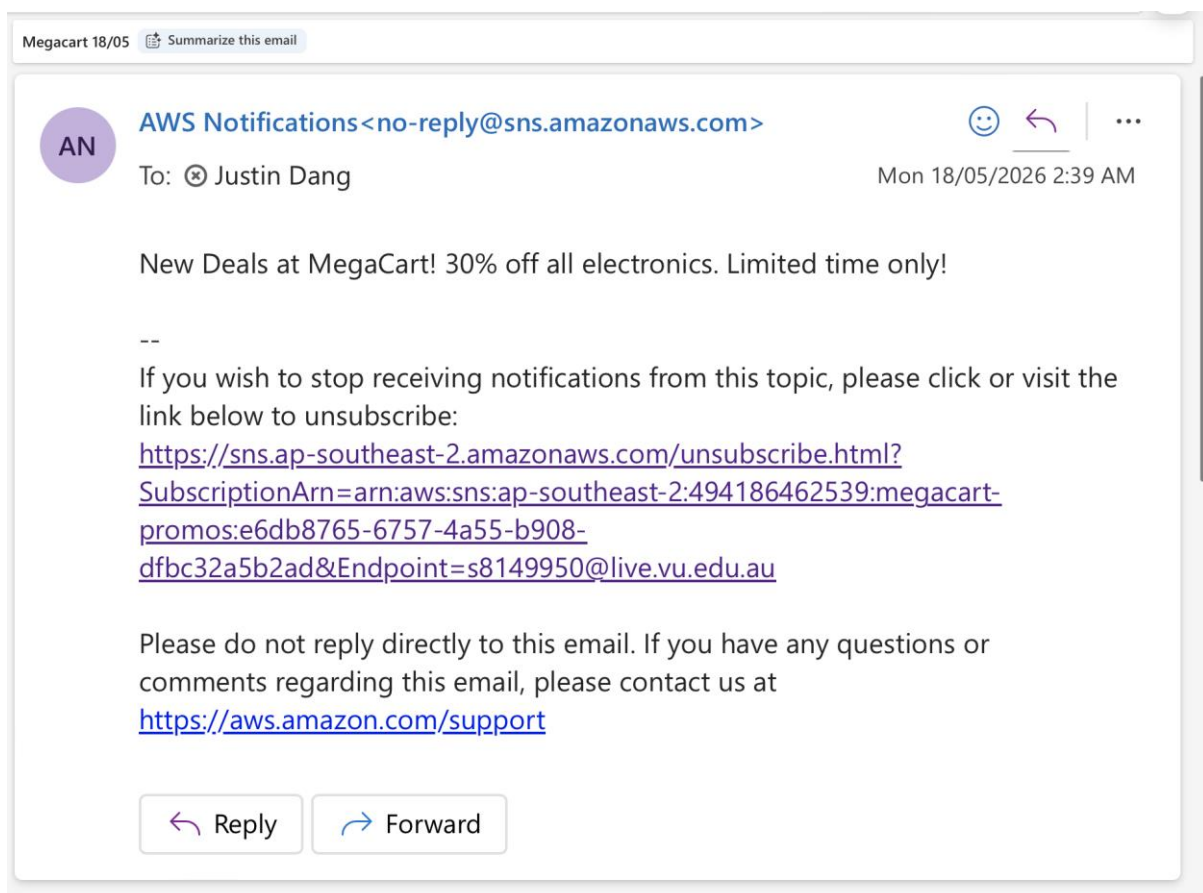
Promotional message successfully delivered to subscribers through Amazon SNS.

Figure 5. MegaCart Website via CloudFront HTTPS



MegaCart website successfully accessed through secure CloudFront HTTPS delivery.

Figure 6. Received Promotional Email Notification



Promotional notification email received from the SNS topic subscribers.

6 Justification of Tools (Incl. AI assistant)

Amazon S3 was used to host the static MegaCart promotional website because it provides a reliable, scalable, and cost-effective storage solution for static web content such as HTML, CSS, JavaScript, and image files. S3 static website hosting is simple to configure and suitable for modern frontend deployment.

Amazon CloudFront was used to securely distribute the website content to users through HTTPS delivery and global edge locations. CloudFront improves performance and scalability by caching website files closer to users, reducing latency and improving loading speed. It also enhances security by enforcing HTTPS connections between users and the website.

Amazon SNS was selected to implement the promotional notification system because it enables fast and reliable message delivery through email or SMS subscriptions. SNS supports real-time communication with customers and can scale efficiently to support large numbers of subscribers.

GitHub Copilot was used as the AI coding assistant during development. It helped generate and improve HTML, CSS, and JavaScript code more efficiently by providing code suggestions, layout ideas, syntax assistance, and faster debugging support. This significantly reduced development time and improved productivity during website creation.

The use of AWS services and GitHub Copilot improved accessibility and ease of deployment by allowing the team to quickly publish, test, and manage the cloud-based solution. AI assistance also supported learning by helping the team understand frontend structure, styling techniques, and coding best practices while building the project.

7 Challenges and Solutions

Table 1. Challenges and Solutions

Challenge	Solution
Issue when opening the website not showing the formatted and stylised website and only showing default text	Checkbox the remaining objects in the bucket that have not been made public.
Unable to collaborate on the same project in AWS	Set IAM user to my teammate
No permission to access AWS services from IAM user	Adding permissions from IAM user groups such as AdministratorAccess and AmazonS3FullAccess

Accidentally created an SNS FIFO topic instead of a Standard topic, causing missing Email/SMS subscription options	Recreated the SNS topic as a Standard topic to enable Email and SMS endpoints
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Summary of implementation issues encountered and their corresponding solutions.

8 Optional Improvements / Next Steps

Several improvements could be added in future development stages to enhance the MegaCart promotional system. Firstly, a custom domain name could be connected to Amazon CloudFront together with an AWS Certificate Manager (ACM) SSL certificate.

Secondly, a CI/CD (Continuous Integration and Continuous Deployment) pipeline could be implemented using GitHub Actions or AWS CodePipeline. This would allow website updates to be automatically deployed to Amazon S3 whenever new code is pushed to the GitHub repository, improving deployment efficiency and reducing manual work.

The notification system could also be expanded by integrating a customer sign-up form into the website, allowing users to subscribe directly to promotional alerts through Amazon SNS.

Finally, future versions of the system could integrate backend services such as AWS Lambda and DynamoDB to support dynamic features including customer accounts, personalised promotions, analytics tracking, and real-time inventory updates.

8.1. CI/CD Pipeline Automation

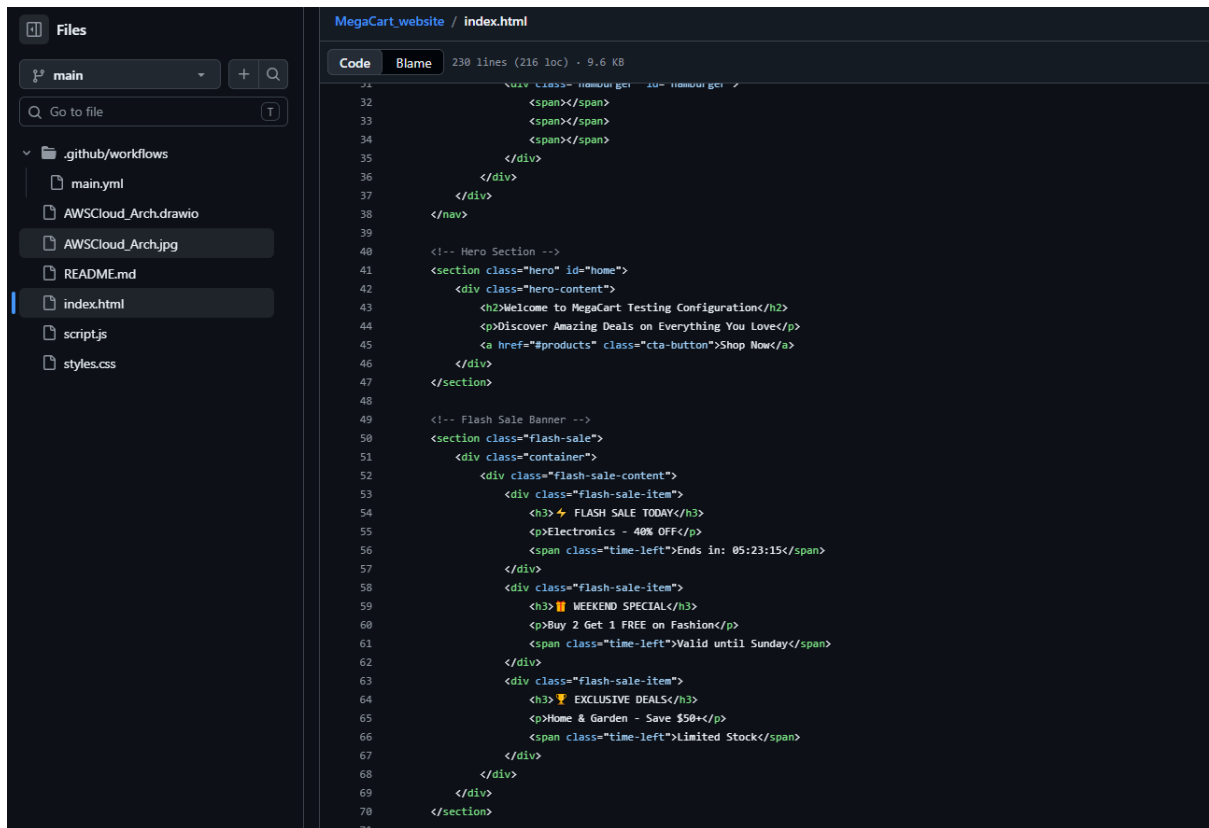
A CI/CD pipeline was successfully implemented using GitHub Actions to automate deployment from GitHub to Amazon S3 and CloudFront. Website updates pushed to the GitHub repository were automatically deployed to AWS, improving deployment efficiency and reducing manual configuration work.

Figure 7

Actions New workflow All workflows Deploy MegaCart website to S3 and Clo... Management Caches Attestations Runners Usage metrics Performance metrics	All workflows Showing runs from all workflows Filter workflow runs 3 workflow runs WorkflowEventStatusBranchActor Update hero section heading for testing configuration Deploy MegaCart website to S3 and CloudFront #3: Commit 4d799c pushed by leonardj-gb 17 minutes ago 32s Delete MegaCart_website/github/workflows directory Deploy MegaCart website to S3 and CloudFront #2: Commit 52f1891 pushed by leonardj-gb 18 minutes ago 17s Add GitHub Actions workflow for S3 deployment Deploy MegaCart website to S3 and CloudFront #1: Commit 43ce2c3 pushed by leonardj-gb 33 minutes ago 17s
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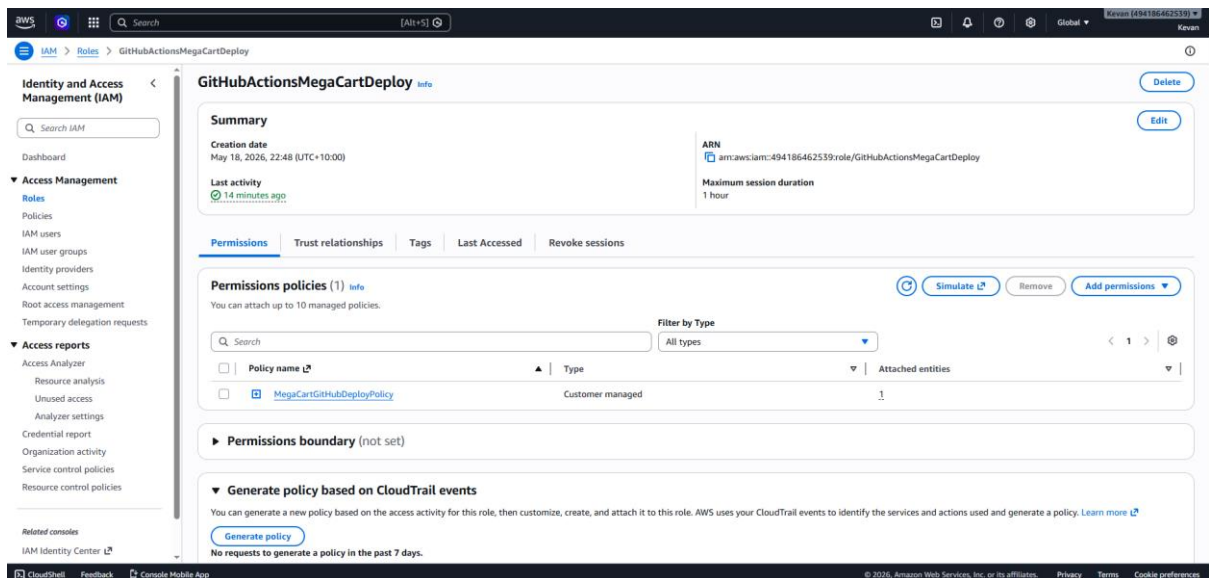
GitHub Actions deployment logs.

Figure 8



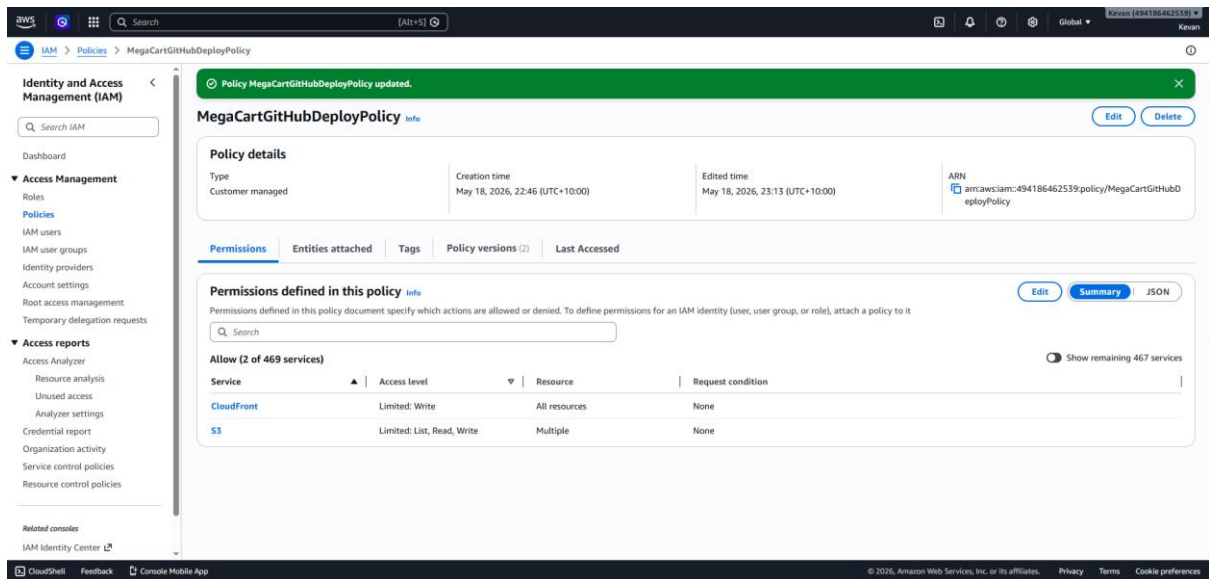
Modified hero section for testing.

Figure 9



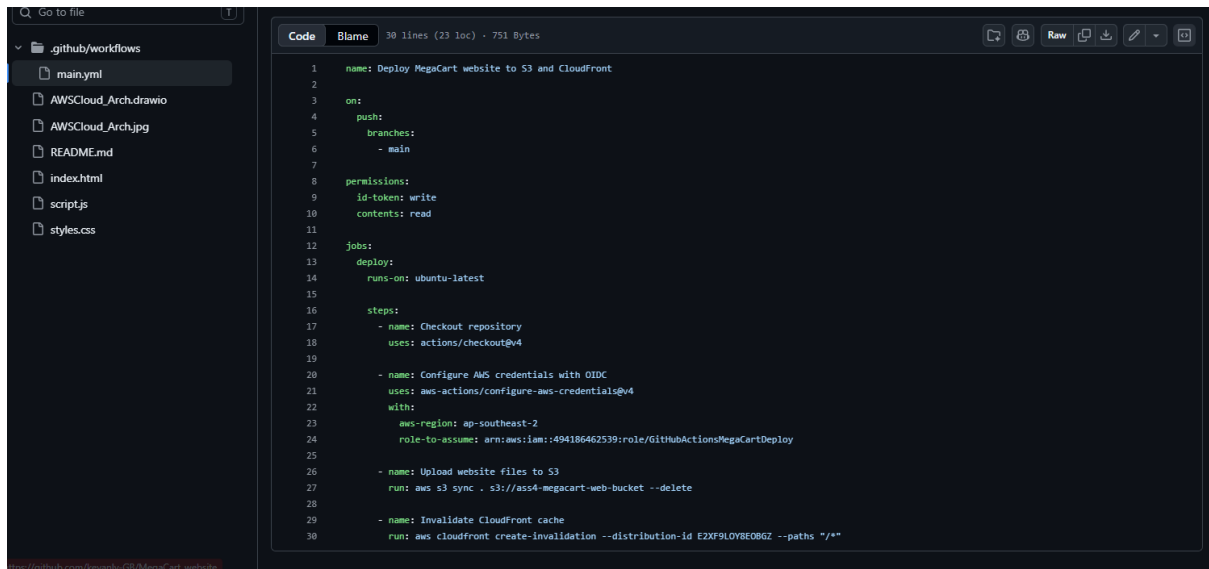
IAM role for GitHub Actions.

Figure 10



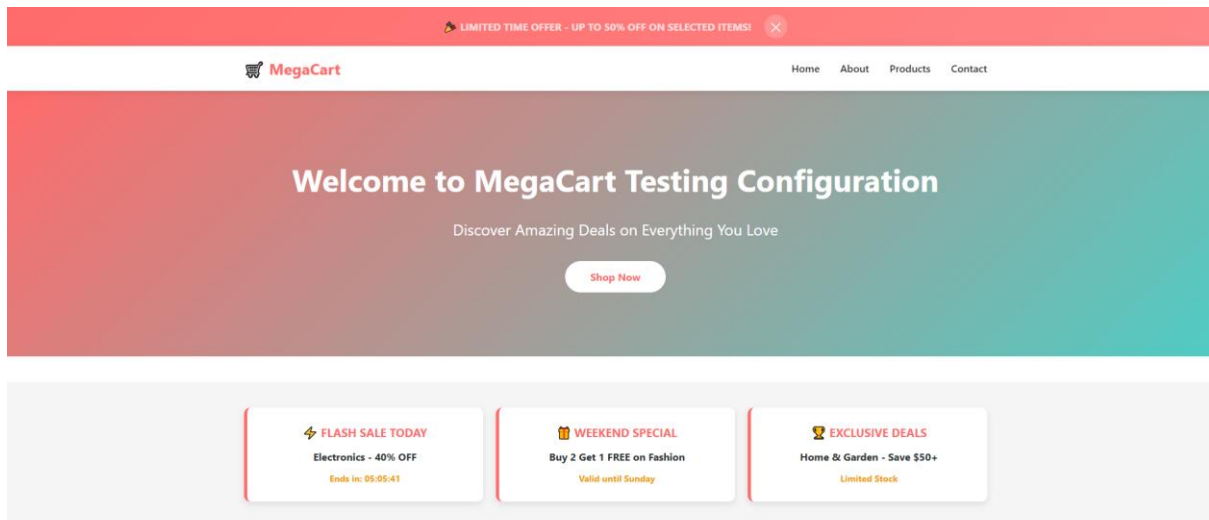
Custom IAM deployment permissions.

Figure 11



GitHub Actions deployment configuration.

Figure 12



Website updated after CI/CD deployment.

9 Reflection on AI Assistant Usage (Github Copilot)

In our project, we have used Github Copilot AI assistant to generate a website for us to fast-track our work to be able to deploy a functional website to host on AWS S3 service. The prompts we specified to the AI were direct in building a website with the required files HTML, CSS and JS, also include features to make the website feel functional and presentable for customers. From thorough inspection of the code files generated, there does not seem to be any pitfall that can be spotted and how we verify this is previewing the website from their console and testing buttons, fields and their responsiveness to cursor movement. Another way to verify the code is through code editor to see how it is structured and follows a coding etiquette.

Reference

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